

Inverses of matrices (contd.)

Theorem If a square matrix has an inverse then it has only one. (If an inverse exists, it is unique.)

Proof Let A be any square matrix.

Suppose B_1 and B_2 are inverses of A .

By associativity,

$$(B_1 A) B_2 = B_1 (A B_2).$$

Because B_1
is inverse of A

Because B_2
is inverse of A

It follows that

$$B_2 = B_1$$



Notation For any invertible matrix A , we denote its (unique) inverse by A^{-1} .

Later: f is a linear transformation

Remark If $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a bijection defined by

$$f(\mathbf{x}) = A\mathbf{x}$$

then the inverse function f^{-1} satisfies

$$f^{-1}(\mathbf{x}) = A^{-1}\mathbf{x}.$$

Theorem Suppose A and B are invertible matrices. Then

$$(AB)^{-1} = B^{-1}A^{-1}$$

in that order!

Proof

We want to show that $B^{-1}A^{-1}$ is the inverse of AB .

Hence we need to check that

$$(AB)(B^{-1}A^{-1}) = I \quad \text{and} \quad (B^{-1}A^{-1})(AB) = I.$$

$$(AB)(B^{-1}A^{-1}) = A(\underbrace{BB^{-1}}_I)A^{-1} = A(\underbrace{IA^{-1}}_{A^{-1}}) = AA^{-1} = I \quad \checkmark$$

associativity

$$(B^{-1}A^{-1})(AB) = I.$$

similarly



This generalizes to more than two matrices:

$$(A_1 A_2 A_3)^{-1} = A_3^{-1} A_2^{-1} A_1^{-1}$$

in this order!

$$(A_1 A_2 \dots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \dots A_1^{-1}.$$

Some logic

Statement: something that is either true or false

Use P , Q , etc. to denote statements

Ex. $P = \text{"11 is prime"}$

$Q = \text{"8 is divisible by 3"}$

Statements can be combined using connectives.

Connective	Symbol
and	$\wedge$
or	$\vee$
if ... then	$\implies$
if and only if (iff)	$\iff$
not	$\neg$

Example If P and Q denote above statements then

$P \wedge Q$ false statement

$P \vee Q$ true statement

$\neg Q$ true statement

$P \wedge \neg Q$ true statement

For any statements P and Q ,

$$\neg(P \wedge Q) = \neg P \vee \neg Q$$

$$\neg(P \vee Q) = \neg P \wedge \neg Q$$

De Morgan's laws

same thing

If ... then

If P then Q

P implies Q

$P \implies Q$

true unless P is true and Q is false

Example "If 11 is prime then 8 is not prime"

11 is prime $\implies$ 8 is not prime
true true

true statement

Example "If 11 is prime then 8 is prime"

11 is prime $\implies$ 8 is prime
true false

false statement

Note If P is false then "If P then Q" ($P \implies Q$)
is true no matter what Q is.

Example 6 is prime $\implies$ 5 is prime

true statement

6 is prime $\implies$ 8 is prime

true statement

P	Q	$P \implies Q$
false	false	true
false	true	true
true	false	false
true	true	true

Definition The converse of $P \implies Q$ is $Q \implies P$.

Definition " $P \iff Q$ " stands for " $P \implies Q$ and $Q \implies P$ "

Remark Definitions are always "if and only if" statements.

Example B is an inverse of A if $AB = BA = I$

read this as "if and only if" ("iff")

Quantifiers

a.) Universal quantifiers

Example Consider the following statement:

For all 3×3 matrices A and B , $AB = BA$.

We can write this statement as follows:

$$\forall A \in M_{3 \times 3}(\mathbb{R}) \quad \forall B \in M_{3 \times 3}(\mathbb{R}), \quad AB = BA$$

The statement is (of course) false.

To show that such a statement is false, one counterexample is enough:

Take for instance

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

Example Consider the following statement:

$$\forall n \in \mathbb{N}, \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

This is a true statement. To prove this, you need to show that

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad \text{holds for any choice of } n \in \mathbb{N}.$$